

15.6

a)

	(1) OLS 1987	(2) OLS 1988
C	0.9348 (0.2010)	0.8993 (0.2407)
EXPER	0.1270 (0.0295)	0.1265 (0.0323)
EXPER ²	-0.0033 (0.0011)	-0.0031 (0.0011)
SOUTH	-0.2128 (0.0338)	-0.2384 (0.0344)
UNION	0.1445 (0.0382)	0.1102 (0.0387)
N	716	716

EXPER is positive on $\ln(WAGE)$ and not change over 2 years.

EXPER² is negative on $\ln(WAGE)$ and not change much over 2 years.

SOUTH is negative and the effect is higher in 1988.

UNION is positive and the effect is lower in 1988.

Assume: - No individual effect: $u_i = 0$, there is no individual heterogeneity

- β_{jit} is the same for all i

b. The $\ln(WAGE)$ equation specified as a panel data regression model is

$$\ln(WAGE_{it}) = \beta_1 + \beta_2 EXPER_{it} + \beta_3 EXPER_{it}^2 + \beta_4 SOUTH_{it} + \beta_5 UNION_{it} + (u_i + e_{it})$$

Explain any differences in assumptions between this model and the models in part (a).

b)

Model includes u_i , capturing the time-invariant individual heterogeneity, so the model allow each woman to have her own individual-specific wage component, like ability, family background, etc.

c)

c. Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?

- EXPER, EXPER² & UNION have smaller effect under FE compared to OLS, this could be because OLS may capture other unobserved individual characteristics.

- south has larger effect in FE, but the standard error is also higher, it could be because the limit of the data, only 2 years.

- The one show the most difference is EXPER (0.12 $\rightarrow$ 0.057)

d)

d. The F -statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F -distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis.

$$F = \frac{(SSE_R - SSE_U) / (N - 1)}{SSE_U / (NT - N - K_3)} = 11.68$$

Null hypothesis:

$$H_0: \beta_{11} = \beta_{12} = \beta_{13} = \dots = \beta_{1N}$$

$$\text{Degree of freedom: } N - 1 = 716 - 1 = 715$$

$$\text{Denominator degree of freedom: } NT - N - K = \overset{(716 \times 2)}{1432} - 716 - 4 = 712$$

$$F(715, 712)$$

The 1% level of significance critical value is: 1.19

Because $11.68 > 1.19 \rightarrow$ reject null hypothesis $\rightarrow$ there are individual difference.

f)

	(4) FE Robust	(5) RE
C	1.5468 (0.2688)	1.1497 (0.1597)
EXPER	0.0575 (0.0328)	0.0986 (0.0220)
EXPER ²	-0.0012 (0.0011)	-0.0023 (0.0007)
SOUTH	-0.3261 (0.2495)	-0.2326 (0.0317)
UNION	0.0822 (0.0367)	0.1027 (0.0245)
N	1432	1432

SOUTH & EXPER coefficients show the most difference.

$$t = \frac{b_{FE,k} - b_{RE,k}}{[\widehat{\text{var}}(b_{FE,k}) - \widehat{\text{var}}(b_{RE,k})]^{1/2}} = \frac{b_{FE,k} - b_{RE,k}}{[se(b_{FE,k})^2 - se(b_{RE,k})^2]^{1/2}}$$

$$t_{EXPER} = \frac{0.0575 - 0.0986}{(0.0328^2 - 0.0220^2)^{1/2}} = -1.67$$

$$t_{SOUTH} = \frac{-0.3261 - (-0.2326)}{(0.2495^2 - 0.0317^2)^{1/2}} = -0.77$$

$$t_{EXPER^2} = \frac{-0.0012 - (-0.0023)}{(0.0011^2 - 0.0007^2)^{1/2}} = 1.3$$

$$t_{UNION} = \frac{0.0822 - 0.1027}{(0.0367^2 - 0.0245^2)^{1/2}} = -1.06$$

$$(-1.67)^2 + (1.3)^2 + (-0.77)^2 + (-1.06)^2 = 6.1954 < \chi^2(4)_{0.05} \approx 9.488$$

→ Fail to reject the Hausman Test

→ Random effect model is appropriate.

15.20

a.

- a. Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	437.76425	1.34622	325.180	< 2e-16	***
small	5.82282	0.98933	5.886	4.19e-09	***
aide	0.81784	0.95299	0.858	0.391	
tchexper	0.49247	0.06956	7.080	1.61e-12	***
boy	-6.15642	0.79613	-7.733	1.23e-14	***
white_asian	3.90581	0.95361	4.096	4.26e-05	***
freelunch	-14.77134	0.89025	-16.592	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 5759 degrees of freedom
(20 observations deleted due to missingness)

Multiple R-squared: 0.09685, Adjusted R-squared: 0.09591

F-statistic: 102.9 on 6 and 5759 DF, p-value: < 2.2e-16

Small class is positively significant on Read score, implying that students in small class perform better in reading,

Aide is positive but insignificant on Read score, meaning that teacher aide has no significant effect on reading score.

Students from experienced teachers have higher scores on reading tests.

Student sex and race have impact on the score. Particularly, boys or white/Asian have lower scores compared to girls.

b.

- b. Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [Hint: specify *SCHID* as the cross-section identifier and *ID* as the "time" identifier.]

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)	
small	6.490231	0.912962	7.1090	1.313e-12	***
aide	0.996087	0.881693	1.1297	0.2586	
tchexper	0.285567	0.070845	4.0309	5.629e-05	***
boy	-5.455941	0.727589	-7.4987	7.440e-14	***
white_asian	8.028019	1.535656	5.2277	1.777e-07	***
freelunch	-14.593572	0.880006	-16.5835	< 2.2e-16	***

Most of the coefficients don't change much, only tchexper and student race, white_asian, increases more than 2 times while tchexper decrease in half.

c.

- c. Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?

> pFtest(fe,mod_a)

F test for individual effects

data: readscore ~ small + aide + tchexper + boy + white_asian + freelunch
F = 16.698, df1 = 78, df2 = 5681, p-value < 2.2e-16
alternative hypothesis: significant effects

Since p-value < 2.2×10^{-16} -> significant -> reject the null hypothesis -> at least one school fixed effect is non zero.

d.

RE

Coefficients:

	Estimate	Std. Error	z-value	Pr(> z)	
(Intercept)	436.126774	2.064782	211.2217	< 2.2e-16	***
small	6.458722	0.912548	7.0777	1.466e-12	***
aide	0.992146	0.881159	1.1260	0.2602	
tchexper	0.302679	0.070292	4.3060	1.662e-05	***
boy	-5.512081	0.727639	-7.5753	3.583e-14	***
white_asian	7.350477	1.431376	5.1353	2.818e-07	***
freelunch	-14.584332	0.874676	-16.6740	< 2.2e-16	***

Compare 3 models:

	OLS	FE	RE
small	5.8228158	6.4902305	6.4587216
aide	0.8178369	0.9960875	0.9921460
tchexper	0.4924687	0.2855668	0.3026787
boy	-6.1564214	-5.4559412	-5.5120812
white_asian	3.9058095	8.0280192	7.3504772
freelunch	-14.7713371	-14.5935724	-14.5843317

Comment: Most of the coefficients don't change much, except of White_asian and tchexper, increasing 2 times in FE and RE and decreasing almost in half, respectively.

The variables most likely to be correlated with school effects are tchexper and white_asian because more experienced teachers may be assigned to certain schools and racial composition may differ systematically across schools.

LM test:

```
> # LM test: H0 = no random effects (Breusch-Pagan LM test)
> plmtest(re, type = "bp")
```

Lagrange Multiplier Test - (Breusch-Pagan)

```
data: readscore ~ small + aide + tchexper + boy + white_asian + freelunch
chisq = 6677.4, df = 1, p-value < 2.2e-16
alternative hypothesis: significant effects
```

Because $p\text{-value} < 0.05 \rightarrow$ reject the null hypothesis $\rightarrow$ school random effects are present.

e.

- e. Using the t -test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?

Hausman Test

```
data: readscore ~ small + aide + tchexper + boy + white_asian + freelunch
chisq = 13.809, df = 6, p-value = 0.03184
alternative hypothesis: one model is inconsistent
```


	b_FE	b_RE	t_stat	reject_H0
small	6.4902	6.4587	1.1460	FALSE
aide	0.9961	0.9921	0.1284	FALSE
tchexper	0.2856	0.3027	-1.9377	FALSE
white_asian	8.0280	7.3505	1.2181	FALSE
freelunch	-14.5936	-14.5843	-0.0956	FALSE

```
> |
```

Fail to reject null hypothesis, there is no strong evidence that FE and RE differ for that coefficient.

```
> t_boy <- (b_fe_boy - b_re_boy) / sqrt(se_fe_boy^2 - se_re_boy^2)
```

Warning message:

In sqrt(se_fe_boy^2 - se_re_boy^2) : NaNs produced

```
> se_fe_boy
[1] 0.7275894
> se_re_boy
[1] 0.7276388
~ |
```

When applying the coefficient-by-coefficient FE versus RE test to BOY, the variance difference is negative. Therefore, the denominator of the test statistic involves the square root of a negative number, and the test cannot be computed. Therefore, based on the Hausman test, we could use the FE.

f.

- f. Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	464.07945	5.47058	84.832	< 2e-16	***
small	6.56177	1.00367	6.538	6.79e-11	***
aide	1.09202	0.96794	1.128	0.259289	
tchexper	0.29460	0.07807	3.774	0.000162	***
boy	-5.40859	0.79969	-6.763	1.48e-11	***
white_asian	8.19637	1.68549	4.863	1.19e-06	***
freelunch	-14.64249	0.96940	-15.105	< 2e-16	***
small_mean	-22.55156	5.82335	-3.873	0.000109	***
aide_mean	10.88891	5.36079	2.031	0.042280	*
tchexper_mean	0.82536	0.18108	4.558	5.28e-06	***
boy_mean	-51.23797	7.16245	-7.154	9.51e-13	***
white_asian_mean	-7.02301	2.35217	-2.986	0.002841	**
freelunch_mean	-3.62902	2.53316	-1.433	0.152025	

Linear hypothesis test:

small_mean = 0

aide_mean = 0

tchexper_mean = 0

boy_mean = 0

white_asian_mean = 0

freelunch_mean = 0

Model 1: restricted model

Model 2: readscore ~ small + aide + tchexper + boy + white_asian + freelunch +
small_mean + aide_mean + tchexper_mean + boy_mean + white_asian_mean +
freelunch_mean

	Res.Df	RSS	Df	Sum of Sq	Chisq	Pr(>Chisq)
1	5695	5220109				
2	5689	5105264	6	114845	127.98	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The Mundlak test p value is small ($<2.2 \times 10^{-16}$) -> reject RE assumption.

The school-average of all variables are significant (except of Freelunch) -> These variables are all correlated to unobserved school-level heterogeneity.

	Estimate	Std. Error	t value	Pr(> t)
small_mean	-22.5515639	5.8233457	-3.872613	1.088825e-04
aide_mean	10.8889073	5.3607855	2.031215	4.227960e-02
tchexper_mean	0.8253595	0.1810846	4.557867	5.275718e-06
boy_mean	-51.2379719	7.1624548	-7.153689	9.513662e-13
white_asian_mean	-7.0230143	2.3521692	-2.985761	2.840786e-03
freelunch_mean	-3.6290153	2.5331563	-1.432606	1.520254e-01